Data scientists build and design complex systems. They analyse and work with large amounts of data and use coding patterns such as python, R and java. A masters degree is a postgraduate degree which involves studying a chosen area in more depth, meaning it gives a better understanding of the subject and develops many needed skills for the role. It is a level 7 qualification and usually lasts one or two years. Data scientists first came about in 1974 when Peter Naur proposed it to replace computer science. A postgraduate degree is not needed to become a data scientist as there are other routes to gaining the skills needed, but a postgraduate qualification gives you a wide range of skills and knowledge needed to be a successful data scientist. Data scientist roles were created to help organisations organise and analyse their data making it easier for them to achieve organisational goals, productivity and decision making.

Data science involves turning raw data into information that is meaningful to organisations to use to improve their business. A data scientist extracts, analyses and interprets large amount of data from a range of sources, they use algorithms, data mining, artificial intelligence, machine learning and statistical tools to make it accessible to businesses. Once the data has been interpreted, the results will be presented using clear and engaging language. Technical, analytical and communication skills are used to collect and examine data to help a business find patterns and solve problems.

Data science is an important field for many sectors, without the knowledge and skills of data scientists workplaces would not be able to increase productivity, organisation and efficiency in the business. Data scientists work in many sectors as

they are required in many workplaces, they work in finance, academia, health care, scientific research, retail, information technology and also for the government. So they work in a range of different sectors. By completing a masters degree it allows you to learn all the skills necessary for the role when you go into the workplace, it teaches you how to use the softwares required for a data scientist to analyse large amounts of data, it also teaches you the skills needed to be a good data scientist and how to go into the workplace with all the knowledge needed to be successful. I think data science is a vital role for organisations across the world, there are many different courses and routes to getting into data science making it easy and flexible to accomplish. They teach you all the skills needed to be successful.

After completing a masters degree in data science it should be expected that many skills are gained to be able to pursue the data scientist role in the workplace. There is a current demand for data scientist so I hope by completing a masters degree I can help contribute towards this shortage and help in the data science sector. Completing this masters degree will put less pressure on the career field as more people come into the role. By taking on this postgraduate degree I expect it to affect the field in a good way, bringing more data scientists out into society, fully qualified to full fill the role in the workplace. I hope to go into the sector to be a data scientist and start gaining hands on experience, and I am hope to learn from the postgraduate degree most of the skills needed to be a successful data scientist and get me into the field. By accomplishing this I hope to contribute to the shortage of data scientists.

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